SECURITIES INDUSTRY AUTOMATION CORPORATION

# National Market Systems

# Common IP Multicast Distribution Network

## Recipient Interface Specification

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| 1.36 | 8/19/10 | MCP | Pages : Appendix ‘A’ (23 & 27)<br>Pages : 8,10 & 11<br><br>IP Source Addresses | - Added OPRA 25-48 HSL’s<br><br>- Revised OPRA “1-24” to 1-48”<br><br>Appendix ‘B’ and Appendix C’ updated to Appendix ‘B’ = Primary Site and ‘C’ for Disaster Recovery IP Source Addresses |
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| 1.40 | 6/25/14 | MCP | Page 9, 11, 12, 15, 21, 25 and 29 | -addition of Extended Session |
| 1.41 | 1/27/15 | LP | Page 23, 27 | -Removed Network A Index references (Indices Disseminated over Network B) |
| 1.42 | 5/4/15 | WG | Page 45 thru 49<br><br><br>Page 50 thru 54 | - Added Appendix C - New Primary Data Center Source Addresses<br>- Added Appendix C - New Disaster Recovery (Backup) Data Center Source Addresses |

| Version | Date | Rev by | Pages affected | Comments |
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| | | | Page 35<br><br><br>Page 36 | - Removed decommissioned source addresses on Network Bor subnets ".41 and .61".<br>- Removed decommissioned source addresses on Network Bor subnets ".42 and .62". |
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| 1.45 | 12/4/20 | WG | Pages 16, 23 and 29<br>Pages 33 thru 47 | -Added new Multicast IP Subnets<br>-Added PILLAR SIP (CTS/CQS) Source and Multicast Addresses and Future OPRA Multicast Addresses |
| 1.46 | 2/24/20 | WG/AE | Appendix C | -Added redundant streams and revised addresses for Pillar SIP Retransmission and Playback test channels |
| 1.47 | 8/6/20 | AE | Whole document | -Removed all references to decommissioned Time Beacon;<br>- Legacy CTA addresses removed |
| 1.48 | 12/16/20 | WG | Pages 16, 23 and 24<br><br>Page 35<br><br>Page 42 | -Added new IP Ranges for OPRA Pillar<br>-Added Source Addresses for OPRA Pillar<br>-Added Appendix D with new Multicast IP Addresses for OPRA Pillar |
| 1.49 | 1/22/21 | WG | Page 45<br>Page 46 | -Corrected Subnet IP Address<br>- Added Missing Subnet IP Address |
| 1.50 | 6/14/21 | WG | Pages 44, 47 and 50<br><br>Pages 51 thru 53 | -Removed inactive IP addresses from OPRA Extended Session<br>-Added Appendix E for new OPRA GTH multicast IP Addresses |
| 1.51 | 1/24/22 | WG | Pages 10, 12, 16, 23, 24, 42 and 43 | -Added references and tables for new Snapshot line and IP Addresses |

| Version | Date | Rev by | Pages affected | Comments |
|---|---|---|---|---|
| | | | | -Removed old OPRA legacy IP Addresses and tables |
| 1.52 | 3/8/23 | WG | Pages 17, 24 and 25<br>Pages 43 thru 65 | -Added New Address Ranges<br>-Added New RP and Source Addresses<br>-Added Appendix 1 thru 5: New OPRA, OPRA GTH, CTS and CQS multicast IP addresses |
| 1.53 | 7/10/23 | WG | Pages 26 thru 65 | -Added decommission and activation dates<br>-Corrected CTA Subnet from /24 to /25<br>-Added multicast line to Source Address table<br>-Corrected port typo for OPRA Retransmission Line 41 and OPRA Playback Test line 42 |
| 1.54 | 9/11/23 | WG | Pages 23 and 24 | Table format change to list old and new IP addresses |
| 1.55 | 2/5/24 | WG | Pages 10, 12 and 13<br><br>Page 16, 26 thru 41<br><br><br>Pages 23 and 24<br><br>Pages 26 thru 43 | Update Number of OPRA Lines and total number of redundancy<br>Removal of old Multicast IPs for subnets; 224.0.86, .88, .89, .90, .92, .93, 94. 95<br>Removal of old CTA and OPRA addresses<br>Move up new multicast IPs for subnets 224.0.200, .201, .202, .203, .204 |
| 1.56 | 5/1/24 | WG | Page 26 | Added OPRA GTH Line References to the Source Subnets table in the “Associated Multicast Lines” column |
| 1.57 | 10/1/24 | CS | Pages 10-16 (former pages 13-26) | -Revised document to reflect that the only direct multicast of NMS Services is via the NMS Network<br>-Deleted as obsolete Section 1.2 “Multicast Primer”; Section 4.1 “IP Multicasting - Primer Part II”; Section 4.3 “UDP/IP Framing”; Section 4.4.4 “Multicast Delivery over ICE Global Network”; Section 4.5 “Logical Groups Mapping versus Physical Access Points”, and Section 5 “Physical, Media Layer, and Network Connectivity” and renumbered remaining Sections |
| 1.58 | 06/13/25 | WG | Pages 37 and 41 | CTS (p.37) and CQS (p.41) Playback Test addresses corrected |
| 1.59 | 02/09/26 | WG | Pages 21, 23, 25, 27, 29 and 31 | Activated Line 220 for OPRA Dynamic Rebalance |

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# 1 Overview

This document provides the interface specifications for customers connecting to the National Market Systems (NMS) distribution network. This includes recipients of the Consolidated Tape System (CTS), Consolidated Quotation System (CQS), and the Options Price Reporting Authority (OPRA) real-time production data. Recipients should also use this specification for information related to receiving NMS real-time data retransmission and NMS after-hours playback data messages. The NMS distribution network disseminates all market data information in the form of multicast addressed IP datagrams.

In order to receive NMS Services (CT/CQ and OPRA), customers must connect to the NMS Network, which is dedicated for NMS Services only.

With respect to physical network connectivity, all data distributed by the NMS systems requires recipients to connect via the NMS Network. Recipients may choose to connect directly to the NMS Network or to connect indirectly using services from ICE Global Network (IGN) or a third-party service provider. For those choosing to connect via IGN, a separate interface specification, "ICE Global Network & Colocation Technical Specifications," should be referenced. For more information regarding the available connectivity options via IGN, please visit the ICE Global Network Website at https://www.theice.com/data-services/global-network, or contact ICE Global Network & Colocation Sales @ 1-770-661-0010 - Option 3 or send an email to iceglobalnetwork-Info@theice.com.

## *1.1 Data Available via the NMS Network*

**NMS Real-Time Production Data**
Two copies (Group A and Group B) of each NMS real-time production messages are available. These redundant copies are delivered via two distinct multicast data streams. For each unique NMS line (**CTS Network A lines 1-12, CTS Network B lines 1-12, CTS Index lines 1-2, CQS Network A lines 1-12, CQS Network B lines 1-12, OPRA lines 1-102 (Regular Session) and OPRA lines 201-204 (Global Trading Hours**) there are two redundant multicast data streams. SIAC refers to these streams as the 'A' and 'B' streams. The 'A' stream is available from Group A and the 'B' stream from Group B. See Appendices A through E for the table of multicast group mappings. Therefore there will be 204 unique multicast groups allocated for the redundant delivery of these 156 lines (156 * 2 =312).

**NMS Real-Time Data Retransmission Data**
The retransmission data streams are available via all network access points and are delivered via two distinct multicast data streams for **CTS Network A lines 1-12, CTS Network B lines 1-12, CTS Index lines 1-2, CQS Network A lines 1-12, CQS Network B lines 1-12, OPRA lines 1-102 (Regular Session) and OPRA lines 201-204 (Global Trading Hours**)**.** The recipient may choose to receive the retransmission data from either or both sites. See Appendices A through E for the table of multicast group mappings. Note: CTS Index lines 1-2 for Snapshot retransmissions is not supported.

**NMS After-Hours Playback Data**
Playback data is available in two methods:

- There is a set of Multicast data feeds dedicated for after-hours playback test data. This playback data is made available via two distinct multicast data streams for **CTS Network A lines 1-12, CTS Network B lines 1-12, CTS Index lines 1-2, CQS Network A lines 1-12, CQS Network B lines 1-12, OPRA lines 1-102 (Regular Session) and OPRA lines 201-204 (Global Trading Hours**)**.** See Appendices A through E for the table of multicast group mappings. Note: CTA After-Hours Playback Data for Snapshot is not supported.

- In addition to the playback test data groups, SIAC will continue to provide dual-sited redundant after-hours playback via the production system expressly for the purposes of redundancy testing.

**Bandwidth Requirements**
Bandwidth requirements change with time and recipients are encouraged to contact the ICE Global Network Help Desk and/or NMS Planning representatives regarding bandwidth requirements of each of the NMS services. For more information on the ICE Global Network and available services, please visit the ICE Global Network Website at https://www.theice.com/data-services/global-network, or contact ICE Global Network & Colocation Sales @ 1-770-661-0010 - Option 3 or send an email to iceglobalnetwork-Info@theice.com.

Recipients should connect to the NMS Network via at least two access points in order to make full use of the redundant data feeds available for each service. Recipients not connecting directly should consult with their provider, whether IGN or a third party, regarding connectivity options.

**Message Formats**
For details of the message formats utilized by the CTS, CQS, and OPRA systems, please reference the following:

- CTS: CTS, Consolidated Tape System, Output Multicast Line, Interface Specification (www.ctaplan.com)
- CQS: CQS, Consolidated Quotation System, Output Multicast Line, Interface Specification (www.ctaplan.com)
- OPRA: OPRA, Options Price Reporting Authority, Data Recipient Interface Specification (www.opraplan.com)

Please note that the message format of retransmission and playback data is also governed by the documents listed above.

The remainder of this specification addresses the communications interfacing requirements for all data types.

Recipients connecting to the NMS Network via IGN are strongly recommended to consult the ICE Global Network and Colocation Technical Specifications, available here: https://www.ice.com/publicdocs/IGN_Colocation_US_Technical_Specifications.pdf, which provides additional information and considerations for receiving multicast services via ICE Global Network.

# 2 NMS Data Types

The NMS network distributes data via the multicast addressing and delivery protocols. Each of the three systems (CTS, CQS, and OPRA) has a unique set of multicast addresses assigned to each of its data “lines”. In each case there are redundant data streams provided for daytime production delivery of each line. The data lines for the three systems include CTS Network A lines 1-12, CTS Network B lines 1-12, CTS Index lines 1-2, CQS Network A lines 1-12, CQS Network B lines 1-12 and OPRA lines 1-102 (Regular Session) and OPRA lines 201-204 (Global Trading Hours). Therefore there will be 312 unique multicast groups allocated for the redundant delivery of these 156 lines (156* 2 =312).

- Each system line has two retransmission multicast address groups assigned to it. Entitled recipients have the option of subscribing to any retransmission line as needed.
- There are additional multicast group ID’s allocated for supporting after-hours playback of the NMS data. Note that after-hours playback can also utilize the aforementioned production multicast groups.
- In all cases, recipients will only be permitted to receive data to which they are entitled.

Appendices A through E provides tables listing of all multicast group ID’s. The tables also include a listing of the UDP destination port numbers assigned to each data stream. The NMS distribution system utilizes the UDP protocol at the IP transport layer. In order to provide the recipient community with the highest level of flexibility, the NMS systems have assigned a unique UDP destination port number to each multicast data stream. Note that the real-time redundant data streams use unique multicast addresses at the IP layer and unique UDP destination port numbers at the transport layer.

Recipient application software may make use of the UDP port mappings in order to multiplex between each of the data streams. Typically, applications use a “socket” programming interface which provides the means for requesting data on a per UDP port basis. If a port is not specified the application host’s operating system might pass all IP broadcast data (including all multicast data) to a single process if the application has not specifically requested data on a per port basis. Please consult your application host’s programming and system documentation for information particular to your environment.

# 3 Application Considerations

This section defines the application data framing and some of the key aspects of the IP distribution environment.

## *3.1 Application Encapsulation*

In the IP environment the NMS application messages are also encapsulated in blocks, which in turn are encapsulated in an Ethernet frame as given in Figure 1 IP Data Block Format.

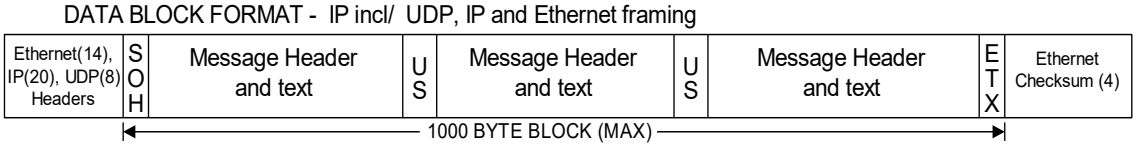



Figure 1 IP Data Block Format

There are several levels of encapsulation that occur within the Ethernet frame. The NMS data block, (which can be a maximum of 1000 bytes), is encapsulated within a UDP datagram, which in turn is encapsulated within an IP datagram which itself is encapsulated within an Ethernet frame. Each number shown in parentheses, e.g., IP (20) refers to the size of the particular header in bytes.

## *3.2 End to End Data Integrity*

Integrity checking, on a per packet basis, is available via a checksum value in both the UDP header (**Error! Reference source not found.**) and the Ethernet frame check sequence.

In general, the Ethernet frame checksum validation is performed by the host's interface firmware and the IP checksum validation is performed within the TCP/IP stack and not by the application software.

Unlike TCP/IP based application services, the UDP/IP protocol has no "built-in" automatic retransmission functionality and therefore recipient host applications must examine the sequence numbers embedded within each NMS message on a per line basis in order to determine whether any data has been missed.

## *3.3 Line Concept*

The term "line" refers to a specific logical data stream identified by the value pair formed by a unique IP multicast destination address and unique UDP destination port number.

Note that the following terms are all analogous to each other:

- multicast group
- multicast group ID
- multicast host group
- multicast host group ID
- multicast destination address

The NMS network currently utilizes 296 unique multicast group ID's for the purposes of providing NMS data to the recipient community. Each multicast group ID also has a UDP destination port number assigned to it, therefore each line of NMS data is uniquely identifiable by the value pair formed by its multicast group and UDP destination port number pair.

Appendices A through E provides the exact mappings of each line to its identifier pair. Reference section 1.2 Multicast Primer for a list summarizing the multicast groups.

# 4 Network Layer Connectivity

## 4.1 Multicast Addressing

The NMS network uses the following ranges, which are presented in further detail starting in Appendices A. Note that not all these addresses are in use.

**SIAC Allocated Multicast IP Address Ranges**

| Starting Address | Ending Address |
|---|---|
| 224.0.200.0 | 224.0.200.255 |
| 224.0.201.0 | 224.0.201.255 |
| 224.0.202.0 | 224.0.202.255 |
| 224.0.203.0 | 224.0.203.255 |
| 224.0.204.0 | 224.0.204.255 |
| 224.0.205.0 | 224.0.205.255 |
| 224.0.206.0 | 224.0.206.255 |
| 224.0.207.0 | 224.0.207.255 |

## 4.2 Multicast Address Use

The multicast group addresses used by SIAC for the dissemination of application data on this network have been registered with the Internet Assigned Numbering Authority (IANA). No recipient will be allowed to connect to the NMS distribution network if it is found that they are using any of these addresses for their own use.

For a list of these addresses please view http://www.iana.org/assignments/multicast-addresses.

### 4.2.1 IGMP

Internet Group Management Protocol (IGMP) is a protocol that end systems use to communicate with multicast compliant routers and is defined in RFC 1112. Recipient host systems that wish to subscribe to multicast groups must be fully compliant with this RFC.

### 4.2.2 Subscription Control

To receive the multicast packets, applications running on recipient end-stations issue IGMP subscription (or “join group”) packets on their locally attached LANs. The local router (which must also be multicast compliant) adds the multicast group to its registration table and begins to forward all packets destined to that group onto the LAN.

Recipients have the option of subscribing to any combination of multicast groups but as mentioned previously, SIAC will allow recipients to receive only those groups to which they have been entitled.

### 4.2.3 Multicast Delivery over NMS Network

NMS Services CT/CQ/OPRA are provided by NMS Network. NMS Network ports are classic Ethernet interface (un-tagged) configured for transporting aggregate multicast traffic for all services. NMS Network lowers the overall latency while maintaining the redundancy and fairness throughout the network.

To facilitate the delivery of Multicast data, NMS Network employs the use of multicast routing protocol. Protocol Independent Multicast (PIM) Sparse mode is utilized to accomplish this task. PIM Sparse mode is used to signal delivery and reception of multicast data. PIM Sparse mode requires the use of Rendezvous Point (RP) to act as meeting point between subscribers and listeners of multicast data. Router behind which interested listeners are, send PIM-Join signal to RP for interested multicast groups to join the Shared Tree for the reception of data. At this stage if the source(s) is already

publishing data and RP has established Multicast Tree towards the Source, the data starts flowing via RP Tree or Shared Tree.

NMS Network customers will have below two methods for receiving multicast data. This specification refers to the configuration of the customer router port directly connected to NMS Network. Customers can implement any network solution they wish beyond that interface. SIAC has no restrictions on the manner in which a customer designs its networks to support multicast reception. This applies to both protocol and physical topology perspectives. Customers are responsible for implementing a working design that best suits their environment.

**Method I: Dynamic Multicast Routing - PIM Sparse Mode**

- **PIM Sparse Mode**: Configure PIM Sparse Mode on the router that connects to Access Network.
- **Rendezvous Point**: RP-A and RP-B IP-Addresses are propagated via dynamic BGP unicast routing for A and B Multicast group mappings.
- **Shared Tree**: Upon receipt of PIM (*, G) Join, Shared Tree is established which is rooted at the Rendezvous Point for Any Source Multicast (ASM) delivery.
- **Shortest Path Tree**: Upon receipt of (S, G) Source / Group Join, Shortest Path Tree (SPT) is established for delivery of Multicast packets.

**Method II: Static Multicast Routing - IGMP Static Groups**

- SIAC, upon the request of the customer, will define IGMP static joins on the NMS Network Edge router connected to the customer. This will result in forwarding of multicast data for the multicast groups that are statically configured. Static IGMP joins are matched against entitled multicast groups to the customer edge router.
- Customer routers learn multicast source routes via dynamic unicast routing as described in section 4.1.1.

Customers can implement appropriate solutions they require on their edge router in order to correctly forward the multicast data into their networks. Typically, router vendors provide the option of forwarding the multicast data at the edge into their routing trees using the routing information learned via dynamic unicast routing.

#### 4.2.3.1 Multicast Entitlement Control

Entitlement for Proprietary Market Data Feeds subscribers will be enforced at the NMS Network Edge Routers through the application of PIM Join Policies on the logical network interface of each individual Customer. The use of Policies provides the mechanism to control the transmission/reception of multicast services for dynamic subscribers. For customers that chose not to use dynamic multicast routing protocols, SIAC will define IGMP static groups for the entitled market data services and applied logical network interface of each individual Customer.

Ingress traffic filters on the Edge Router logical interfaces supporting multicast will silently discard any incoming packets except those used by the required multicast (PIM) and/or unicast routing protocols (BGP). These filters will also be used to protect each of the network components within the NMS Network from any customer-originated multicast traffic. Filters can be reconfigured dynamically to allow for timely re-provisioning of entitlements.

#### 4.2.3.2 Multicast Data Retransmission

NMS Multicast data service provides an in-band retransmission request mechanism over a unicast TCP sockets based applications. NMS Multicast data service supports multicast retransmission over separate group ranges as defined in Appendix-A of this document. NMS Network subscribers are able to receive multicast retransmission over the same physical and/or logical interface as their primary production feeds.

Customers can implement any network solution they wish beyond the NMS Network interface. Beyond the NMS Network demarcation point, SIAC places no restrictions on the manner in which a customer designs its networks to support multicast reception. This is true from both from a protocol and physical topology perspectives. Customers are responsible for implementing a working design that best suits their environments.

## *4.3 Data Entitlement*

For a recipient host system to receive a particular data stream it must subscribe to the data stream's corresponding multicast group ID via IGMP. Appendices A through C lists all multicast group ID assignments.

In order to restrict a recipient from subscribing to data streams that they are not entitled to, outbound packet filters are employed on SIAC's distribution routers interfaces connecting to the recipients. These filters block data from being sent to non-entitled recipients on a per service basis (CTS, CQS, and OPRA).

## *4.4 IP Addressing Considerations*

Please consult the ICE Global Network interface specification for details.

## *4.5 Recipient Security*

SIAC protects its network and hosts using several methods. Traffic filters and routing policies prevent sharing of information and data between entities connected to the NMS Network. Additional measures are in place as well, however these security measures maintain the integrity of SIAC's distribution environment by protecting SIAC's network and hosts from intentional or accidental access from within a recipient network.

These measures are in no way intended to provide the same level of security to the recipients themselves. If a recipient believes that additional security is required to protect their network they are encouraged to take action to implement additional security measures.

For the purposes of aiding in the implementation of security measures (e.g. traffic filters), the source IP addresses associated with the NMS systems have been provided in Appendices C and D.

# 5 Appendix A - NMS IP Multicast Addresses

This appendix contains the mapping of IP multicast group ID's (addresses) to the currently available data lines. To receive a particular data stream the recipient host system would typically subscribe to that particular multicast group ID. Two multicast group ID's are available for each real-time production data line. The data originating from Group A is generally referred to as the 'A' streams and the data from Group B as the 'B' streams. Also provided in the table are the UDP destination ports associated with each logical line.

The NMS data messages are encapsulated in an identical manner in both streams. For example, a datagram issued Group A on OPRA Line 2 destined to multicast group 224.0.204.2 will have a corresponding datagram (containing the identical UDP data payload, i.e. same NMS messages and same sequence number range) sourced from Group B destined to multicast group 224.0.206.2.

**Multicast Address Ranges:**

| NMS PRODUCTION IP MULTICAST FEEDS GROUP A |
|---|
| 224.0.200.0 - 224.0.200.15 |
| 224.0.200.96 - 224.0.200.99 |
| 224.0.202.0 - 224.0.202.31 |
| 224.0.202.32 - 224.0.202.63 |
| 224.0.203.0 - 224.0.203.31 |
| 224.0.203.32 - 224.0.203.63 |
| 224.0.204.0 - 224.0.204.127 |

| NMS PRODUCTION IP MULTICAST FEEDS GROUP B |
|---|
| 224.0.200.128 - 224.0.200.143 |
| 224.0.200.224 - 224.0.200.227 |
| 224.0.202.128 - 224.0.202.159 |
| 224.0.202.160 - 224.0.202.191 |
| 224.0.203.128 - 224.0.203.159 |
| 224.0.203.160 - 224.0.203.191 |
| 224.0.206.0 - 224.0.206.127 |

# Appendix A - NMS IP Multicast Addresses, Cont'd

**Retransmission and Playback Test Data, Single and Dual Sets**

CTA supports dual data stream sets for the CTS and CQS retransmission and playback test data feeds (Appendix C). OPRA, supports dual data stream sets for retransmission and playback test data feeds (Appendix D). Playback data is only available after-hours.

Recipients wishing to receive retransmission and/or playback feeds must subscribe to the multicast feeds based on the addressing information shown in the following table.

**Multicast Address Ranges:**

| NMS PRODUCTION IP MULTICAST FEEDS GROUP A |
|---|
| |
| **RETRANSMISSION GROUPS** |
| 224.0.200.100 - 224.0.200.103 |
| 224.0.200.16 - 224.0.200.31 |
| 224.0.201.0 - 224.0.201.31 |
| 224.0.201.32 - 224.0.201.63 |
| 224.0.202.64 - 224.0.202.95 |
| 224.0.202.96 - 224.0.202.127 |
| 224.0.203.64 - 224.0.203.95 |
| 224.0.203.96 - 224.0.203.127 |
| 224.0.204.128 - 224.0.204.255 |
| |
| **PLAYBACK TEST GROUPS** |
| 224.0.200.104 - 224.0.200.107 |
| 224.0.200.32 - 224.0.200.47 |
| 224.0.200.56 - 224.0.200.57 |
| 224.0.200.58 - 224.0.200.59 |
| 224.0.200.60 - 224.0.200.61 |
| 224.0.200.62 - 224.0.200.63 |
| 224.0.205.0 - 224.0.205.127 |

| NMS PRODUCTION IP MULTICAST FEEDS GROUP B |
|---|
| |
| **RETRANSMISSIONS GROUPS** |
| 224.0.200.228 - 224.0.200.231 |
| 224.0.200.144 - 224.0.200.159 |
| 224.0.201.128 - 224.0.201.159 |
| 224.0.201.160 - 224.0.201.191 |
| 224.0.202.192 - 224.0.202.223 |
| 224.0.202.224 - 224.0.202.255 |
| 224.0.203.192 - 224.0.203.223 |
| 224.0.203.224 - 224.0.203.255 |
| 224.0.206.128 - 224.0.206.255 |
| |
| **PLAYBACK TEST GROUPS** |
| 224.0.200.160 - 224.0.200.175 |
| 224.0.200.184 - 224.0.200.185 |
| 224.0.200.186 - 224.0.200.187 |
| 224.0.200.188 - 224.0.200.189 |
| 224.0.200.190 - 224.0.200.191 |
| 224.0.200.232 - 224.0.200.235 |
| 224.0.207.0 - 224.0.207.127 |

# 6 Appendix 1 - Rendezvous Points and SIP Data Center Source Addresses

**RENDEZVOUS POINTS FOR NMS NETWORK**

| NETWORK CONNECTION | RP Address - A STREAM | RP Address - B STREAM |
|---|---|---|
| NMS Network | 159.125.57.222 | 159.125.57.223 |

**MULTICAST DATA SOURCE SUBNETS FOR PRIMARY AND DISASTER DATA CENTERS**

| SYSTEM / DATA STREAM | PRIMARY SOURCE IP/SUBNET | DISASTER SOURCE IP/SUBNET | ASSOCIATED MULTICAST LINES |
|---|---|---|---|
| CTS & CQS Multicast A1 | 159.125.48.0/25 | 198.140.48.0/25 | DATA STREAM A<br>TAPE A LINES 1-12 |
| CTS & CQS Multicast A2 | 159.125.48.128/25 | 198.140.48.128/25 | DATA STREAM A<br>TAPE B LINES 1-12 &<br>INDEX LINES 1-2 |
| | | | |
| CTS & CQS Multicast B1 | 159.125.49.0/25 | 198.140.49.0/25 | DATA STREAM B<br>TAPE A LINES 1-12 |
| CTS & CQS Multicast B2 | 159.125.49.128/25 | 198.140.49.128/25 | DATA STREAM B<br>TAPE B LINES 1-12 &<br>INDEX LINES 1-2 |
| | | | |
| OPRA Multicast A1 | 162.69.44.0/24 | 162.68.44.0/24 | DATA STREAM A<br>OPRA LINES 1 - 24<br>OPRA GTH LINE 1 |
| OPRA Multicast A2 | 162.69.45.0/24 | 162.68.45.0/24 | DATA STREAM A<br>OPRA LINES 25 - 48<br>OPRA GTH LINE 2 |
| OPRA Multicast A3 | 162.69.46.0/24 | 162.68.46.0/24 | DATA STREAM A<br>OPRA LINES 49 - 72<br>OPRA GTH LINE 3 |
| OPRA Multicast A4 | 162.69.47.0/24 | 162.68.47.0/24 | DATA STREAM A<br>OPRA LINES 73 - 96 &<br>RESERVED LINES 220 - 225<br>OPRA GTH LINE 4 |
| | | | |
| OPRA Multicast B1 | 162.69.52.0/24 | 162.68.52.0/24 | DATA STREAM B<br>OPRA LINES 1 - 24<br>OPRA GTH LINE 1 |
| OPRA Multicast B2 | 162.69.53.0/24 | 162.68.53.0/24 | DATA STREAM B<br>OPRA LINES 25 - 48<br>OPRA GTH LINE 2 |
| OPRA Multicast B3 | 162.69.54.0/24 | 162.68.54.0/24 | DATA STREAM B<br>OPRA LINES 49 - 72<br>OPRA GTH LINE 3 |
| OPRA Multicast B4 | 162.69.55.0/24 | 162.68.55.0/24 | DATA STREAM B<br>OPRA LINES 73 - 96 &<br>RESERVED LINES 220 - 225<br>OPRA GTH LINE 4 |
| | | | |

**Note: Multicast Addresses can be referenced on the next following pages.**

# 7 Appendix 2 - OPRA Regular Session Production, Real-Time IP Multicast Feeds, (Network 'A' only)

| PRODUCT NAME: | | OPRA Regular Session | | | |
|---|---|---|---|---|---|
| Network Subnet: | | ONE (1) NETWORK SUBNET GROUP PER DATA STREAM | | | |
| **PRODUCTION "DATA STREAM - A" SUBNET: 224.0.204.0/25** | | | | | |
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| OPRA 1 | 224.0.204.1 | 45001 | OPRA 49 | 224.0.204.49 | 45049 |
| OPRA 2 | 224.0.204.2 | 45002 | OPRA 50 | 224.0.204.50 | 45050 |
| OPRA 3 | 224.0.204.3 | 45003 | OPRA 51 | 224.0.204.51 | 45051 |
| OPRA 4 | 224.0.204.4 | 45004 | OPRA 52 | 224.0.204.52 | 45052 |
| OPRA 5 | 224.0.204.5 | 45005 | OPRA 53 | 224.0.204.53 | 45053 |
| OPRA 6 | 224.0.204.6 | 45006 | OPRA 54 | 224.0.204.54 | 45054 |
| OPRA 7 | 224.0.204.7 | 45007 | OPRA 55 | 224.0.204.55 | 45055 |
| OPRA 8 | 224.0.204.8 | 45008 | OPRA 56 | 224.0.204.56 | 45056 |
| OPRA 9 | 224.0.204.9 | 45009 | OPRA 57 | 224.0.204.57 | 45057 |
| OPRA 10 | 224.0.204.10 | 45010 | OPRA 58 | 224.0.204.58 | 45058 |
| OPRA 11 | 224.0.204.11 | 45011 | OPRA 59 | 224.0.204.59 | 45059 |
| OPRA 12 | 224.0.204.12 | 45012 | OPRA 60 | 224.0.204.60 | 45060 |
| OPRA 13 | 224.0.204.13 | 45013 | OPRA 61 | 224.0.204.61 | 45061 |
| OPRA 14 | 224.0.204.14 | 45014 | OPRA 62 | 224.0.204.62 | 45062 |
| OPRA 15 | 224.0.204.15 | 45015 | OPRA 63 | 224.0.204.63 | 45063 |
| OPRA 16 | 224.0.204.16 | 45016 | OPRA 64 | 224.0.204.64 | 45064 |
| OPRA 17 | 224.0.204.17 | 45017 | OPRA 65 | 224.0.204.65 | 45065 |
| OPRA 18 | 224.0.204.18 | 45018 | OPRA 66 | 224.0.204.66 | 45066 |
| OPRA 19 | 224.0.204.19 | 45019 | OPRA 67 | 224.0.204.67 | 45067 |
| OPRA 20 | 224.0.204.20 | 45020 | OPRA 68 | 224.0.204.68 | 45068 |
| OPRA 21 | 224.0.204.21 | 45021 | OPRA 69 | 224.0.204.69 | 45069 |
| OPRA 22 | 224.0.204.22 | 45022 | OPRA 70 | 224.0.204.70 | 45070 |
| OPRA 23 | 224.0.204.23 | 45023 | OPRA 71 | 224.0.204.71 | 45071 |
| OPRA 24 | 224.0.204.24 | 45024 | OPRA 72 | 224.0.204.72 | 45072 |
| OPRA 25 | 224.0.204.25 | 45025 | OPRA 73 | 224.0.204.73 | 45073 |
| OPRA 26 | 224.0.204.26 | 45026 | OPRA 74 | 224.0.204.74 | 45074 |
| OPRA 27 | 224.0.204.27 | 45027 | OPRA 75 | 224.0.204.75 | 45075 |
| OPRA 28 | 224.0.204.28 | 45028 | OPRA 76 | 224.0.204.76 | 45076 |
| OPRA 29 | 224.0.204.29 | 45029 | OPRA 77 | 224.0.204.77 | 45077 |
| OPRA 30 | 224.0.204.30 | 45030 | OPRA 78 | 224.0.204.78 | 45078 |
| OPRA 31 | 224.0.204.31 | 45031 | OPRA 79 | 224.0.204.79 | 45079 |
| OPRA 32 | 224.0.204.32 | 45032 | OPRA 80 | 224.0.204.80 | 45080 |
| OPRA 33 | 224.0.204.33 | 45033 | OPRA 81 | 224.0.204.81 | 45081 |
| OPRA 34 | 224.0.204.34 | 45034 | OPRA 82 | 224.0.204.82 | 45082 |
| OPRA 35 | 224.0.204.35 | 45035 | OPRA 83 | 224.0.204.83 | 45083 |
| OPRA 36 | 224.0.204.36 | 45036 | OPRA 84 | 224.0.204.84 | 45084 |
| OPRA 37 | 224.0.204.37 | 45037 | OPRA 85 | 224.0.204.85 | 45085 |
| OPRA 38 | 224.0.204.38 | 45038 | OPRA 86 | 224.0.204.86 | 45086 |
| OPRA 39 | 224.0.204.39 | 45039 | OPRA 87 | 224.0.204.87 | 45087 |
| OPRA 40 | 224.0.204.40 | 45040 | OPRA 88 | 224.0.204.88 | 45088 |
| OPRA 41 | 224.0.204.41 | 45041 | OPRA 89 | 224.0.204.89 | 45089 |
| OPRA 42 | 224.0.204.42 | 45042 | OPRA 90 | 224.0.204.90 | 45090 |
| OPRA 43 | 224.0.204.43 | 45043 | OPRA 91 | 224.0.204.91 | 45091 |
| OPRA 44 | 224.0.204.44 | 45044 | OPRA 92 | 224.0.204.92 | 45092 |
| OPRA 45 | 224.0.204.45 | 45045 | OPRA 93 | 224.0.204.93 | 45093 |
| OPRA 46 | 224.0.204.46 | 45046 | OPRA 94 | 224.0.204.94 | 45094 |
| OPRA 47 | 224.0.204.47 | 45047 | OPRA 95 | 224.0.204.95 | 45095 |
| OPRA 48 | 224.0.204.48 | 45048 | OPRA 96 | 224.0.204.96 | 45096 |

# Appendix 2 - OPRA Regular Session Production, Real-Time IP Multicast Feeds, (Network 'A' only) - CONTINUED

| PRODUCT NAME: | OPRA Regular Session |
|---|---|

**PRODUCTION "DATA STREAM - A" SUBNET: 224.0.204.0/25**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 220 | 224.0.204.105 | 45105 |
| **OPRA 221 | 224.0.204.106 | 45106 |
| **OPRA 222 | 224.0.204.107 | 45107 |
| **OPRA 223 | 224.0.204.108 | 45108 |
| **OPRA 224 | 224.0.204.109 | 45109 |
| **OPRA 225 | 224.0.204.110 | 45110 |

** RESERVED FOR FUTURE USE

Note: OPRA 220 will be used with the OPRA Daily Symbol Rebalance

# Appendix 2 - OPRA Regular Session Production, Real-Time IP Multicast Feeds, (Network 'B' only)

| PRODUCT NAME: | OPRA Regular Session | | | | |
|---|---|---|---|---|---|
| Network Subnet: | ONE (1) NETWORK SUBNET GROUP PER DATA STREAM | | | | |
| **PRODUCTION "DATA STREAM - B" SUBNET: 224.0.206.0/25** | | | | | |
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| OPRA 1 | 224.0.206.1 | 45001 | OPRA 49 | 224.0.206.49 | 45049 |
| OPRA 2 | 224.0.206.2 | 45002 | OPRA 50 | 224.0.206.50 | 45050 |
| OPRA 3 | 224.0.206.3 | 45003 | OPRA 51 | 224.0.206.51 | 45051 |
| OPRA 4 | 224.0.206.4 | 45004 | OPRA 52 | 224.0.206.52 | 45052 |
| OPRA 5 | 224.0.206.5 | 45005 | OPRA 53 | 224.0.206.53 | 45053 |
| OPRA 6 | 224.0.206.6 | 45006 | OPRA 54 | 224.0.206.54 | 45054 |
| OPRA 7 | 224.0.206.7 | 45007 | OPRA 55 | 224.0.206.55 | 45055 |
| OPRA 8 | 224.0.206.8 | 45008 | OPRA 56 | 224.0.206.56 | 45056 |
| OPRA 9 | 224.0.206.9 | 45009 | OPRA 57 | 224.0.206.57 | 45057 |
| OPRA 10 | 224.0.206.10 | 45010 | OPRA 58 | 224.0.206.58 | 45058 |
| OPRA 11 | 224.0.206.11 | 45011 | OPRA 59 | 224.0.206.59 | 45059 |
| OPRA 12 | 224.0.206.12 | 45012 | OPRA 60 | 224.0.206.60 | 45060 |
| OPRA 13 | 224.0.206.13 | 45013 | OPRA 61 | 224.0.206.61 | 45061 |
| OPRA 14 | 224.0.206.14 | 45014 | OPRA 62 | 224.0.206.62 | 45062 |
| OPRA 15 | 224.0.206.15 | 45015 | OPRA 63 | 224.0.206.63 | 45063 |
| OPRA 16 | 224.0.206.16 | 45016 | OPRA 64 | 224.0.206.64 | 45064 |
| OPRA 17 | 224.0.206.17 | 45017 | OPRA 65 | 224.0.206.65 | 45065 |
| OPRA 18 | 224.0.206.18 | 45018 | OPRA 66 | 224.0.206.66 | 45066 |
| OPRA 19 | 224.0.206.19 | 45019 | OPRA 67 | 224.0.206.67 | 45067 |
| OPRA 20 | 224.0.206.20 | 45020 | OPRA 68 | 224.0.206.68 | 45068 |
| OPRA 21 | 224.0.206.21 | 45021 | OPRA 69 | 224.0.206.69 | 45069 |
| OPRA 22 | 224.0.206.22 | 45022 | OPRA 70 | 224.0.206.70 | 45070 |
| OPRA 23 | 224.0.206.23 | 45023 | OPRA 71 | 224.0.206.71 | 45071 |
| OPRA 24 | 224.0.206.24 | 45024 | OPRA 72 | 224.0.206.72 | 45072 |
| OPRA 25 | 224.0.206.25 | 45025 | OPRA 73 | 224.0.206.73 | 45073 |
| OPRA 26 | 224.0.206.26 | 45026 | OPRA 74 | 224.0.206.74 | 45074 |
| OPRA 27 | 224.0.206.27 | 45027 | OPRA 75 | 224.0.206.75 | 45075 |
| OPRA 28 | 224.0.206.28 | 45028 | OPRA 76 | 224.0.206.76 | 45076 |
| OPRA 29 | 224.0.206.29 | 45029 | OPRA 77 | 224.0.206.77 | 45077 |
| OPRA 30 | 224.0.206.30 | 45030 | OPRA 78 | 224.0.206.78 | 45078 |
| OPRA 31 | 224.0.206.31 | 45031 | OPRA 79 | 224.0.206.79 | 45079 |
| OPRA 32 | 224.0.206.32 | 45032 | OPRA 80 | 224.0.206.80 | 45080 |
| OPRA 33 | 224.0.206.33 | 45033 | OPRA 81 | 224.0.206.81 | 45081 |
| OPRA 34 | 224.0.206.34 | 45034 | OPRA 82 | 224.0.206.82 | 45082 |
| OPRA 35 | 224.0.206.35 | 45035 | OPRA 83 | 224.0.206.83 | 45083 |
| OPRA 36 | 224.0.206.36 | 45036 | OPRA 84 | 224.0.206.84 | 45084 |
| OPRA 37 | 224.0.206.37 | 45037 | OPRA 85 | 224.0.206.85 | 45085 |
| OPRA 38 | 224.0.206.38 | 45038 | OPRA 86 | 224.0.206.86 | 45086 |
| OPRA 39 | 224.0.206.39 | 45039 | OPRA 87 | 224.0.206.87 | 45087 |
| OPRA 40 | 224.0.206.40 | 45040 | OPRA 88 | 224.0.206.88 | 45088 |
| OPRA 41 | 224.0.206.41 | 45041 | OPRA 89 | 224.0.206.89 | 45089 |
| OPRA 42 | 224.0.206.42 | 45042 | OPRA 90 | 224.0.206.90 | 45090 |
| OPRA 43 | 224.0.206.43 | 45043 | OPRA 91 | 224.0.206.91 | 45091 |
| OPRA 44 | 224.0.206.44 | 45044 | OPRA 92 | 224.0.206.92 | 45092 |
| OPRA 45 | 224.0.206.45 | 45045 | OPRA 93 | 224.0.206.93 | 45093 |
| OPRA 46 | 224.0.206.46 | 45046 | OPRA 94 | 224.0.206.94 | 45094 |
| OPRA 47 | 224.0.206.47 | 45047 | OPRA 95 | 224.0.206.95 | 45095 |
| OPRA 48 | 224.0.206.48 | 45048 | OPRA 96 | 224.0.206.96 | 45096 |

# Appendix 2 - OPRA Regular Session Production, Real-Time IP Multicast Feeds, (Network 'B' only) - CONTINUED

| PRODUCT NAME: | OPRA Regular Session |
|---|---|

**PRODUCTION "DATA STREAM - B" SUBNET: 224.0.206.0/25**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 220 | 224.0.206.105 | 45105 |
| **OPRA 221 | 224.0.206.106 | 45106 |
| **OPRA 222 | 224.0.206.107 | 45107 |
| **OPRA 223 | 224.0.206.108 | 45108 |
| **OPRA 224 | 224.0.206.109 | 45109 |
| **OPRA 225 | 224.0.206.110 | 45110 |

** RESERVED FOR FUTURE USE

Note: OPRA 220 will be used with the OPRA Daily Symbol Rebalance

# Appendix 2 - OPRA Regular Session Retransmission IP Multicast Feeds, (Network 'A' only)

| PRODUCT NAME: | OPRA Regular Session | | | | |
|---|---|---|---|---|---|
| Network Subnet: | ONE (1) NETWORK SUBNET GROUP PER DATA STREAM | | | | |
| **RETRANSMISSION "DATA STREAM - A" SUBNET: 224.0.204.128/25** | | | | | |
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| OPRA 1 | 224.0.204.128 | 46001 | OPRA 49 | 224.0.204.176 | 46049 |
| OPRA 2 | 224.0.204.129 | 46002 | OPRA 50 | 224.0.204.177 | 46050 |
| OPRA 3 | 224.0.204.130 | 46003 | OPRA 51 | 224.0.204.178 | 46051 |
| OPRA 4 | 224.0.204.131 | 46004 | OPRA 52 | 224.0.204.179 | 46052 |
| OPRA 5 | 224.0.204.132 | 46005 | OPRA 53 | 224.0.204.180 | 46053 |
| OPRA 6 | 224.0.204.133 | 46006 | OPRA 54 | 224.0.204.181 | 46054 |
| OPRA 7 | 224.0.204.134 | 46007 | OPRA 55 | 224.0.204.182 | 46055 |
| OPRA 8 | 224.0.204.135 | 46008 | OPRA 56 | 224.0.204.183 | 46056 |
| OPRA 9 | 224.0.204.136 | 46009 | OPRA 57 | 224.0.204.184 | 46057 |
| OPRA 10 | 224.0.204.137 | 46010 | OPRA 58 | 224.0.204.185 | 46058 |
| OPRA 11 | 224.0.204.138 | 46011 | OPRA 59 | 224.0.204.186 | 46059 |
| OPRA 12 | 224.0.204.139 | 46012 | OPRA 60 | 224.0.204.187 | 46060 |
| OPRA 13 | 224.0.204.140 | 46013 | OPRA 61 | 224.0.204.188 | 46061 |
| OPRA 14 | 224.0.204.141 | 46014 | OPRA 62 | 224.0.204.189 | 46062 |
| OPRA 15 | 224.0.204.142 | 46015 | OPRA 63 | 224.0.204.190 | 46063 |
| OPRA 16 | 224.0.204.143 | 46016 | OPRA 64 | 224.0.204.191 | 46064 |
| OPRA 17 | 224.0.204.144 | 46017 | OPRA 65 | 224.0.204.192 | 46065 |
| OPRA 18 | 224.0.204.145 | 46018 | OPRA 66 | 224.0.204.193 | 46066 |
| OPRA 19 | 224.0.204.146 | 46019 | OPRA 67 | 224.0.204.194 | 46067 |
| OPRA 20 | 224.0.204.147 | 46020 | OPRA 68 | 224.0.204.195 | 46068 |
| OPRA 21 | 224.0.204.148 | 46021 | OPRA 69 | 224.0.204.196 | 46069 |
| OPRA 22 | 224.0.204.149 | 46022 | OPRA 70 | 224.0.204.197 | 46070 |
| OPRA 23 | 224.0.204.150 | 46023 | OPRA 71 | 224.0.204.198 | 46071 |
| OPRA 24 | 224.0.204.151 | 46024 | OPRA 72 | 224.0.204.199 | 46072 |
| OPRA 25 | 224.0.204.152 | 46025 | OPRA 73 | 224.0.204.200 | 46073 |
| OPRA 26 | 224.0.204.153 | 46026 | OPRA 74 | 224.0.204.201 | 46074 |
| OPRA 27 | 224.0.204.154 | 46027 | OPRA 75 | 224.0.204.202 | 46075 |
| OPRA 28 | 224.0.204.155 | 46028 | OPRA 76 | 224.0.204.203 | 46076 |
| OPRA 29 | 224.0.204.156 | 46029 | OPRA 77 | 224.0.204.204 | 46077 |
| OPRA 30 | 224.0.204.157 | 46030 | OPRA 78 | 224.0.204.205 | 46078 |
| OPRA 31 | 224.0.204.158 | 46031 | OPRA 79 | 224.0.204.206 | 46079 |
| OPRA 32 | 224.0.204.159 | 46032 | OPRA 80 | 224.0.204.207 | 46080 |
| OPRA 33 | 224.0.204.160 | 46033 | OPRA 81 | 224.0.204.208 | 46081 |
| OPRA 34 | 224.0.204.161 | 46034 | OPRA 82 | 224.0.204.209 | 46082 |
| OPRA 35 | 224.0.204.162 | 46035 | OPRA 83 | 224.0.204.210 | 46083 |
| OPRA 36 | 224.0.204.163 | 46036 | OPRA 84 | 224.0.204.211 | 46084 |
| OPRA 37 | 224.0.204.164 | 46037 | OPRA 85 | 224.0.204.212 | 46085 |
| OPRA 38 | 224.0.204.165 | 46038 | OPRA 86 | 224.0.204.213 | 46086 |
| OPRA 39 | 224.0.204.166 | 46039 | OPRA 87 | 224.0.204.214 | 46087 |
| OPRA 40 | 224.0.204.167 | 46040 | OPRA 88 | 224.0.204.215 | 46088 |
| OPRA 41 | 224.0.204.168 | 46041 | OPRA 89 | 224.0.204.216 | 46089 |
| OPRA 42 | 224.0.204.169 | 46042 | OPRA 90 | 224.0.204.217 | 46090 |
| OPRA 43 | 224.0.204.170 | 46043 | OPRA 91 | 224.0.204.218 | 46091 |
| OPRA 44 | 224.0.204.171 | 46044 | OPRA 92 | 224.0.204.219 | 46092 |
| OPRA 45 | 224.0.204.172 | 46045 | OPRA 93 | 224.0.204.220 | 46093 |
| OPRA 46 | 224.0.204.173 | 46046 | OPRA 94 | 224.0.204.221 | 46094 |
| OPRA 47 | 224.0.204.174 | 46047 | OPRA 95 | 224.0.204.222 | 46095 |
| OPRA 48 | 224.0.204.175 | 46048 | OPRA 96 | 224.0.204.223 | 46096 |

# Appendix 2 - OPRA Regular Session Retransmission IP Multicast Feeds, (Network 'A' only) - CONTINUED

| PRODUCT NAME: | OPRA Regular Session |
|---|---|

**RETRANSMISSION "DATA STREAM - A" SUBNET: 224.0.204.128/25**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 220 | 224.0.204.232 | 46232 |
| **OPRA 221 | 224.0.204.233 | 46233 |
| **OPRA 222 | 224.0.204.234 | 46234 |
| **OPRA 223 | 224.0.204.235 | 46235 |
| **OPRA 224 | 224.0.204.236 | 46236 |
| **OPRA 225 | 224.0.204.237 | 46237 |

** RESERVED FOR FUTURE USE

Note: OPRA 220 will be used with the OPRA Daily Symbol Rebalance

# Appendix 2 - OPRA Regular Session Retransmission IP Multicast Feeds, (Network 'B' only)

| PRODUCT NAME: | OPRA Regular Session | | | | |
|---|---|---|---|---|---|
| Network Subnet: | ONE (1) NETWORK SUBNET GROUP PER DATA STREAM | | | | |
| **RETRANSMISSION "DATA STREAM - B" SUBNET: 224.0.206.128/25** | | | | | |
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| OPRA 1 | 224.0.206.128 | 46001 | OPRA 49 | 224.0.206.176 | 46049 |
| OPRA 2 | 224.0.206.129 | 46002 | OPRA 50 | 224.0.206.177 | 46050 |
| OPRA 3 | 224.0.206.130 | 46003 | OPRA 51 | 224.0.206.178 | 46051 |
| OPRA 4 | 224.0.206.131 | 46004 | OPRA 52 | 224.0.206.179 | 46052 |
| OPRA 5 | 224.0.206.132 | 46005 | OPRA 53 | 224.0.206.180 | 46053 |
| OPRA 6 | 224.0.206.133 | 46006 | OPRA 54 | 224.0.206.181 | 46054 |
| OPRA 7 | 224.0.206.134 | 46007 | OPRA 55 | 224.0.206.182 | 46055 |
| OPRA 8 | 224.0.206.135 | 46008 | OPRA 56 | 224.0.206.183 | 46056 |
| OPRA 9 | 224.0.206.136 | 46009 | OPRA 57 | 224.0.206.184 | 46057 |
| OPRA 10 | 224.0.206.137 | 46010 | OPRA 58 | 224.0.206.185 | 46058 |
| OPRA 11 | 224.0.206.138 | 46011 | OPRA 59 | 224.0.206.186 | 46059 |
| OPRA 12 | 224.0.206.139 | 46012 | OPRA 60 | 224.0.206.187 | 46060 |
| OPRA 13 | 224.0.206.140 | 46013 | OPRA 61 | 224.0.206.188 | 46061 |
| OPRA 14 | 224.0.206.141 | 46014 | OPRA 62 | 224.0.206.189 | 46062 |
| OPRA 15 | 224.0.206.142 | 46015 | OPRA 63 | 224.0.206.190 | 46063 |
| OPRA 16 | 224.0.206.143 | 46016 | OPRA 64 | 224.0.206.191 | 46064 |
| OPRA 17 | 224.0.206.144 | 46017 | OPRA 65 | 224.0.206.192 | 46065 |
| OPRA 18 | 224.0.206.145 | 46018 | OPRA 66 | 224.0.206.193 | 46066 |
| OPRA 19 | 224.0.206.146 | 46019 | OPRA 67 | 224.0.206.194 | 46067 |
| OPRA 20 | 224.0.206.147 | 46020 | OPRA 68 | 224.0.206.195 | 46068 |
| OPRA 21 | 224.0.206.148 | 46021 | OPRA 69 | 224.0.206.196 | 46069 |
| OPRA 22 | 224.0.206.149 | 46022 | OPRA 70 | 224.0.206.197 | 46070 |
| OPRA 23 | 224.0.206.150 | 46023 | OPRA 71 | 224.0.206.198 | 46071 |
| OPRA 24 | 224.0.206.151 | 46024 | OPRA 72 | 224.0.206.199 | 46072 |
| OPRA 25 | 224.0.206.152 | 46025 | OPRA 73 | 224.0.206.200 | 46073 |
| OPRA 26 | 224.0.206.153 | 46026 | OPRA 74 | 224.0.206.201 | 46074 |
| OPRA 27 | 224.0.206.154 | 46027 | OPRA 75 | 224.0.206.202 | 46075 |
| OPRA 28 | 224.0.206.155 | 46028 | OPRA 76 | 224.0.206.203 | 46076 |
| OPRA 29 | 224.0.206.156 | 46029 | OPRA 77 | 224.0.206.204 | 46077 |
| OPRA 30 | 224.0.206.157 | 46030 | OPRA 78 | 224.0.206.205 | 46078 |
| OPRA 31 | 224.0.206.158 | 46031 | OPRA 79 | 224.0.206.206 | 46079 |
| OPRA 32 | 224.0.206.159 | 46032 | OPRA 80 | 224.0.206.207 | 46080 |
| OPRA 33 | 224.0.206.160 | 46033 | OPRA 81 | 224.0.206.208 | 46081 |
| OPRA 34 | 224.0.206.161 | 46034 | OPRA 82 | 224.0.206.209 | 46082 |
| OPRA 35 | 224.0.206.162 | 46035 | OPRA 83 | 224.0.206.210 | 46083 |
| OPRA 36 | 224.0.206.163 | 46036 | OPRA 84 | 224.0.206.211 | 46084 |
| OPRA 37 | 224.0.206.164 | 46037 | OPRA 85 | 224.0.206.212 | 46085 |
| OPRA 38 | 224.0.206.165 | 46038 | OPRA 86 | 224.0.206.213 | 46086 |
| OPRA 39 | 224.0.206.166 | 46039 | OPRA 87 | 224.0.206.214 | 46087 |
| OPRA 40 | 224.0.206.167 | 46040 | OPRA 88 | 224.0.206.215 | 46088 |
| OPRA 41 | 224.0.206.168 | 46041 | OPRA 89 | 224.0.206.216 | 46089 |
| OPRA 42 | 224.0.206.169 | 46042 | OPRA 90 | 224.0.206.217 | 46090 |
| OPRA 43 | 224.0.206.170 | 46043 | OPRA 91 | 224.0.206.218 | 46091 |
| OPRA 44 | 224.0.206.171 | 46044 | OPRA 92 | 224.0.206.219 | 46092 |
| OPRA 45 | 224.0.206.172 | 46045 | OPRA 93 | 224.0.206.220 | 46093 |
| OPRA 46 | 224.0.206.173 | 46046 | OPRA 94 | 224.0.206.221 | 46094 |
| OPRA 47 | 224.0.206.174 | 46047 | OPRA 95 | 224.0.206.222 | 46095 |
| OPRA 48 | 224.0.206.175 | 46048 | OPRA 96 | 224.0.206.223 | 46096 |

# Appendix 2 - OPRA Regular Session Retransmission IP Multicast Feeds, (Network 'B' only) - CONTINUED

| PRODUCT NAME: | OPRA Regular Session |
|---|---|

**RETRANSMISSION "DATA STREAM - B" SUBNET: 224.0.206.128/25**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 220 | 224.0.206.232 | 46232 |
| **OPRA 221 | 224.0.206.233 | 46233 |
| **OPRA 222 | 224.0.206.234 | 46234 |
| **OPRA 223 | 224.0.206.235 | 46235 |
| **OPRA 224 | 224.0.206.236 | 46236 |
| **OPRA 225 | 224.0.206.237 | 46237 |

** RESERVED FOR FUTURE USE

Note: OPRA 220 will be used with the OPRA Daily Symbol Rebalance

# Appendix 2 - OPRA Regular Session Playback Test IP Multicast Feeds, (Network 'A' only)

| PRODUCT NAME: | | OPRA Regular Session | | | |
|---|---|---|---|---|---|
| Network Subnet: | | ONE (1) NETWORK SUBNET GROUP PER DATA STREAM | | | |
| **PLAYBACK TEST "DATA STREAM - A" SUBNET: 224.0.205.0/25** | | | | | |
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| OPRA 1 | 224.0.205.1 | 47001 | OPRA 49 | 224.0.205.49 | 47049 |
| OPRA 2 | 224.0.205.2 | 47002 | OPRA 50 | 224.0.205.50 | 47050 |
| OPRA 3 | 224.0.205.3 | 47003 | OPRA 51 | 224.0.205.51 | 47051 |
| OPRA 4 | 224.0.205.4 | 47004 | OPRA 52 | 224.0.205.52 | 47052 |
| OPRA 5 | 224.0.205.5 | 47005 | OPRA 53 | 224.0.205.53 | 47053 |
| OPRA 6 | 224.0.205.6 | 47006 | OPRA 54 | 224.0.205.54 | 47054 |
| OPRA 7 | 224.0.205.7 | 47007 | OPRA 55 | 224.0.205.55 | 47055 |
| OPRA 8 | 224.0.205.8 | 47008 | OPRA 56 | 224.0.205.56 | 47056 |
| OPRA 9 | 224.0.205.9 | 47009 | OPRA 57 | 224.0.205.57 | 47057 |
| OPRA 10 | 224.0.205.10 | 47010 | OPRA 58 | 224.0.205.58 | 47058 |
| OPRA 11 | 224.0.205.11 | 47011 | OPRA 59 | 224.0.205.59 | 47059 |
| OPRA 12 | 224.0.205.12 | 47012 | OPRA 60 | 224.0.205.60 | 47060 |
| OPRA 13 | 224.0.205.13 | 47013 | OPRA 61 | 224.0.205.61 | 47061 |
| OPRA 14 | 224.0.205.14 | 47014 | OPRA 62 | 224.0.205.62 | 47062 |
| OPRA 15 | 224.0.205.15 | 47015 | OPRA 63 | 224.0.205.63 | 47063 |
| OPRA 16 | 224.0.205.16 | 47016 | OPRA 64 | 224.0.205.64 | 47064 |
| OPRA 17 | 224.0.205.17 | 47017 | OPRA 65 | 224.0.205.65 | 47065 |
| OPRA 18 | 224.0.205.18 | 47018 | OPRA 66 | 224.0.205.66 | 47066 |
| OPRA 19 | 224.0.205.19 | 47019 | OPRA 67 | 224.0.205.67 | 47067 |
| OPRA 20 | 224.0.205.20 | 47020 | OPRA 68 | 224.0.205.68 | 47068 |
| OPRA 21 | 224.0.205.21 | 47021 | OPRA 69 | 224.0.205.69 | 47069 |
| OPRA 22 | 224.0.205.22 | 47022 | OPRA 70 | 224.0.205.70 | 47070 |
| OPRA 23 | 224.0.205.23 | 47023 | OPRA 71 | 224.0.205.71 | 47071 |
| OPRA 24 | 224.0.205.24 | 47024 | OPRA 72 | 224.0.205.72 | 47072 |
| OPRA 25 | 224.0.205.25 | 47025 | OPRA 73 | 224.0.205.73 | 47073 |
| OPRA 26 | 224.0.205.26 | 47026 | OPRA 74 | 224.0.205.74 | 47074 |
| OPRA 27 | 224.0.205.27 | 47027 | OPRA 75 | 224.0.205.75 | 47075 |
| OPRA 28 | 224.0.205.28 | 47028 | OPRA 76 | 224.0.205.76 | 47076 |
| OPRA 29 | 224.0.205.29 | 47029 | OPRA 77 | 224.0.205.77 | 47077 |
| OPRA 30 | 224.0.205.30 | 47030 | OPRA 78 | 224.0.205.78 | 47078 |
| OPRA 31 | 224.0.205.31 | 47031 | OPRA 79 | 224.0.205.79 | 47079 |
| OPRA 32 | 224.0.205.32 | 47032 | OPRA 80 | 224.0.205.80 | 47080 |
| OPRA 33 | 224.0.205.33 | 47033 | OPRA 81 | 224.0.205.81 | 47081 |
| OPRA 34 | 224.0.205.34 | 47034 | OPRA 82 | 224.0.205.82 | 47082 |
| OPRA 35 | 224.0.205.35 | 47035 | OPRA 83 | 224.0.205.83 | 47083 |
| OPRA 36 | 224.0.205.36 | 47036 | OPRA 84 | 224.0.205.84 | 47084 |
| OPRA 37 | 224.0.205.37 | 47037 | OPRA 85 | 224.0.205.85 | 47085 |
| OPRA 38 | 224.0.205.38 | 47038 | OPRA 86 | 224.0.205.86 | 47086 |
| OPRA 39 | 224.0.205.39 | 47039 | OPRA 87 | 224.0.205.87 | 47087 |
| OPRA 40 | 224.0.205.40 | 47040 | OPRA 88 | 224.0.205.88 | 47088 |
| OPRA 41 | 224.0.205.41 | 47041 | OPRA 89 | 224.0.205.89 | 47089 |
| OPRA 42 | 224.0.205.42 | 47042 | OPRA 90 | 224.0.205.90 | 47090 |
| OPRA 43 | 224.0.205.43 | 47043 | OPRA 91 | 224.0.205.91 | 47091 |
| OPRA 44 | 224.0.205.44 | 47044 | OPRA 92 | 224.0.205.92 | 47092 |
| OPRA 45 | 224.0.205.45 | 47045 | OPRA 93 | 224.0.205.93 | 47093 |
| OPRA 46 | 224.0.205.46 | 47046 | OPRA 94 | 224.0.205.94 | 47094 |
| OPRA 47 | 224.0.205.47 | 47047 | OPRA 95 | 224.0.205.95 | 47095 |
| OPRA 48 | 224.0.205.48 | 47048 | OPRA 96 | 224.0.205.96 | 47096 |

# Appendix 2 - OPRA Regular Session Playback Test IP Multicast Feeds, (Network 'A' only) - CONTINUED

| PRODUCT NAME: | OPRA Regular Session | |
|---|---|---|
| | | |
| **PLAYBACK TEST "DATA STREAM - A" SUBNET: 224.0.205.0/25** | | |
| | | |
| Originated Data Line | Multicast Group Address | Multicast Group Port |
| OPRA 220 | 224.0.205.105 | 47105 |
| **OPRA 221 | 224.0.205.106 | 47106 |
| **OPRA 222 | 224.0.205.107 | 47107 |
| **OPRA 223 | 224.0.205.108 | 47108 |
| **OPRA 224 | 224.0.205.109 | 47109 |
| **OPRA 225 | 224.0.205.110 | 47110 |

** RESERVED FOR FUTURE USE

Note: OPRA 220 will be used with the OPRA Daily Symbol Rebalance

# Appendix 2 - OPRA Regular Session Playback Test IP Multicast Feeds, (Network 'B' only)

| PRODUCT NAME: | | OPRA Regular Session | | | |
|---|---|---|---|---|---|
| Network Subnet: | | ONE (1) NETWORK SUBNET GROUP PER DATA STREAM | | | |
| **PLAYBACK TEST "DATA STREAM - B" SUBNET: 224.0.207.0/25** | | | | | |
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| OPRA 1 | 224.0.207.1 | 47001 | OPRA 49 | 224.0.207.49 | 47049 |
| OPRA 2 | 224.0.207.2 | 47002 | OPRA 50 | 224.0.207.50 | 47050 |
| OPRA 3 | 224.0.207.3 | 47003 | OPRA 51 | 224.0.207.51 | 47051 |
| OPRA 4 | 224.0.207.4 | 47004 | OPRA 52 | 224.0.207.52 | 47052 |
| OPRA 5 | 224.0.207.5 | 47005 | OPRA 53 | 224.0.207.53 | 47053 |
| OPRA 6 | 224.0.207.6 | 47006 | OPRA 54 | 224.0.207.54 | 47054 |
| OPRA 7 | 224.0.207.7 | 47007 | OPRA 55 | 224.0.207.55 | 47055 |
| OPRA 8 | 224.0.207.8 | 47008 | OPRA 56 | 224.0.207.56 | 47056 |
| OPRA 9 | 224.0.207.9 | 47009 | OPRA 57 | 224.0.207.57 | 47057 |
| OPRA 10 | 224.0.207.10 | 47010 | OPRA 58 | 224.0.207.58 | 47058 |
| OPRA 11 | 224.0.207.11 | 47011 | OPRA 59 | 224.0.207.59 | 47059 |
| OPRA 12 | 224.0.207.12 | 47012 | OPRA 60 | 224.0.207.60 | 47060 |
| OPRA 13 | 224.0.207.13 | 47013 | OPRA 61 | 224.0.207.61 | 47061 |
| OPRA 14 | 224.0.207.14 | 47014 | OPRA 62 | 224.0.207.62 | 47062 |
| OPRA 15 | 224.0.207.15 | 47015 | OPRA 63 | 224.0.207.63 | 47063 |
| OPRA 16 | 224.0.207.16 | 47016 | OPRA 64 | 224.0.207.64 | 47064 |
| OPRA 17 | 224.0.207.17 | 47017 | OPRA 65 | 224.0.207.65 | 47065 |
| OPRA 18 | 224.0.207.18 | 47018 | OPRA 66 | 224.0.207.66 | 47066 |
| OPRA 19 | 224.0.207.19 | 47019 | OPRA 67 | 224.0.207.67 | 47067 |
| OPRA 20 | 224.0.207.20 | 47020 | OPRA 68 | 224.0.207.68 | 47068 |
| OPRA 21 | 224.0.207.21 | 47021 | OPRA 69 | 224.0.207.69 | 47069 |
| OPRA 22 | 224.0.207.22 | 47022 | OPRA 70 | 224.0.207.70 | 47070 |
| OPRA 23 | 224.0.207.23 | 47023 | OPRA 71 | 224.0.207.71 | 47071 |
| OPRA 24 | 224.0.207.24 | 47024 | OPRA 72 | 224.0.207.72 | 47072 |
| OPRA 25 | 224.0.207.25 | 47025 | OPRA 73 | 224.0.207.73 | 47073 |
| OPRA 26 | 224.0.207.26 | 47026 | OPRA 74 | 224.0.207.74 | 47074 |
| OPRA 27 | 224.0.207.27 | 47027 | OPRA 75 | 224.0.207.75 | 47075 |
| OPRA 28 | 224.0.207.28 | 47028 | OPRA 76 | 224.0.207.76 | 47076 |
| OPRA 29 | 224.0.207.29 | 47029 | OPRA 77 | 224.0.207.77 | 47077 |
| OPRA 30 | 224.0.207.30 | 47030 | OPRA 78 | 224.0.207.78 | 47078 |
| OPRA 31 | 224.0.207.31 | 47031 | OPRA 79 | 224.0.207.79 | 47079 |
| OPRA 32 | 224.0.207.32 | 47032 | OPRA 80 | 224.0.207.80 | 47080 |
| OPRA 33 | 224.0.207.33 | 47033 | OPRA 81 | 224.0.207.81 | 47081 |
| OPRA 34 | 224.0.207.34 | 47034 | OPRA 82 | 224.0.207.82 | 47082 |
| OPRA 35 | 224.0.207.35 | 47035 | OPRA 83 | 224.0.207.83 | 47083 |
| OPRA 36 | 224.0.207.36 | 47036 | OPRA 84 | 224.0.207.84 | 47084 |
| OPRA 37 | 224.0.207.37 | 47037 | OPRA 85 | 224.0.207.85 | 47085 |
| OPRA 38 | 224.0.207.38 | 47038 | OPRA 86 | 224.0.207.86 | 47086 |
| OPRA 39 | 224.0.207.39 | 47039 | OPRA 87 | 224.0.207.87 | 47087 |
| OPRA 40 | 224.0.207.40 | 47040 | OPRA 88 | 224.0.207.88 | 47088 |
| OPRA 41 | 224.0.207.41 | 47041 | OPRA 89 | 224.0.207.89 | 47089 |
| OPRA 42 | 224.0.207.42 | 47042 | OPRA 90 | 224.0.207.90 | 47090 |
| OPRA 43 | 224.0.207.43 | 47043 | OPRA 91 | 224.0.207.91 | 47091 |
| OPRA 44 | 224.0.207.44 | 47044 | OPRA 92 | 224.0.207.92 | 47092 |
| OPRA 45 | 224.0.207.45 | 47045 | OPRA 93 | 224.0.207.93 | 47093 |
| OPRA 46 | 224.0.207.46 | 47046 | OPRA 94 | 224.0.207.94 | 47094 |
| OPRA 47 | 224.0.207.47 | 47047 | OPRA 95 | 224.0.207.95 | 47095 |
| OPRA 48 | 224.0.207.48 | 47048 | OPRA 96 | 224.0.207.96 | 47096 |

# Appendix 2 - OPRA Regular Session Playback Test IP Multicast Feeds, (Network 'B' only) - CONTINUED

| PRODUCT NAME: | OPRA Regular Session |
|---|---|

**PLAYBACK TEST "DATA STREAM - B" SUBNET: 224.0.207.0/25**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 220 | 224.0.207.105 | 47105 |
| OPRA 221 | 224.0.207.106 | 47106 |
| OPRA 222 | 224.0.207.107 | 47107 |
| OPRA 223 | 224.0.207.108 | 47108 |
| OPRA 224 | 224.0.207.109 | 47109 |
| OPRA 225 | 224.0.207.110 | 47110 |

** RESERVED FOR FUTURE USE

Note: OPRA 220 will be used with the OPRA Daily Symbol Rebalance

# 8 Appendix 3 - OPRA Global Trading Hours (GTH) Production, Real-Time IP Multicast Feeds, Dual Sets (Network ‘A’ / ‘B’ Data Streams)

| PRODUCT NAME: | OPRA Global Trading Hours (GTH) |
|---|---|
| NETWORK SUBNETS: | ONE (1) NETWORK SUBNET GROUP PER DATA STREAM |

**PRODUCTION “DATA STREAM - A” SUBNET: 224.0.200.0/28**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 201 | 224.0.200.1 | 48001 |
| OPRA 202 | 224.0.200.2 | 48002 |
| OPRA 203 | 224.0.200.3 | 48003 |
| OPRA 204 | 224.0.200.4 | 48004 |

**PRODUCTION “DATA STREAM - B” SUBNET: 224.0.200.128/28**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 201 | 224.0.200.128 | 48001 |
| OPRA 202 | 224.0.200.129 | 48002 |
| OPRA 203 | 224.0.200.130 | 48003 |
| OPRA 204 | 224.0.200.131 | 48004 |

# Appendix 3 - OPRA Global Trading Hours (GTH) Retransmission IP Multicast Feeds, Dual Sets (Network 'A' / 'B' Data Streams)

| PRODUCT NAME: | OPRA Global Trading Hours (GTH) |
|---|---|
| NETWORK SUBNETS: | ONE (1) NETWORK SUBNET GROUP PER DATA STREAM |

**RETRANSMISSION "DATA STREAM - A" SUBNET: 224.0.200.16/28**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 201 | 224.0.200.16 | 49001 |
| OPRA 202 | 224.0.200.17 | 49002 |
| OPRA 203 | 224.0.200.18 | 49003 |
| OPRA 204 | 224.0.200.19 | 49004 |

**RETRANSMISSION "DATA STREAM - B" SUBNET: 224.0.200.144/28**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 201 | 224.0.200.144 | 49001 |
| OPRA 202 | 224.0.200.145 | 49002 |
| OPRA 203 | 224.0.200.146 | 49003 |
| OPRA 204 | 224.0.200.147 | 49004 |

# Appendix 3 - OPRA Global Trading Hours (GTH) Playback Test IP Multicast Feeds, Dual Sets (Network 'A' / 'B' Data Streams)

| PRODUCT NAME: | OPRA Global Trading Hours (GTH) |
|---|---|
| NETWORK SUBNETS: | ONE (1) NETWORK SUBNET GROUP PER DATA STREAM |

**PLAYBACK TEST "DATA STREAM - A" SUBNET: 224.0.200.32/28**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 201 | 224.0.200.32 | 50001 |
| OPRA 202 | 224.0.200.33 | 50002 |
| OPRA 203 | 224.0.200.34 | 50003 |
| OPRA 204 | 224.0.200.35 | 50004 |

**PLAYBACK TEST "DATA STREAM - B" SUBNET: 224.0.200.160/28**

| Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|
| OPRA 201 | 224.0.200.160 | 50001 |
| OPRA 202 | 224.0.200.161 | 50002 |
| OPRA 203 | 224.0.200.162 | 50003 |
| OPRA 204 | 224.0.200.163 | 50004 |

# 9 Appendix 4 - CQS Production, Real-Time IP Multicast Feeds, Dual Sets (Network 'A' / Network 'B')

**MULTICAST DATA: PRODUCTION "A-STREAM" & PRODUCTION "B-STREAM"**

| PRODUCT NAME: | CQS |
|---|---|
| NETWORK SUBNETS: | TWO (2) NETWORK SUBNETS GROUPS PER DATA STREAM |

**PRODUCTION "DATA STREAM - A" SUBNETS:**

| 224.0.203.0/28 | | | 224.0.203.32/28 | | |
|---|---|---|---|---|---|
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| CQS 1 / Tape A | 224.0.203.1 | 45001 | CQS 1 / Tape B | 224.0.203.32 | 45001 |
| CQS 2 / Tape A | 224.0.203.2 | 45002 | CQS 2 / Tape B | 224.0.203.33 | 45002 |
| CQS 3 / Tape A | 224.0.203.3 | 45003 | CQS 3 / Tape B | 224.0.203.34 | 45003 |
| CQS 4 / Tape A | 224.0.203.4 | 45004 | CQS 4 / Tape B | 224.0.203.35 | 45004 |
| CQS 5 / Tape A | 224.0.203.5 | 45005 | CQS 5 / Tape B | 224.0.203.36 | 45005 |
| CQS 6 / Tape A | 224.0.203.6 | 45006 | CQS 6 / Tape B | 224.0.203.37 | 45006 |
| CQS 7 / Tape A | 224.0.203.7 | 45007 | CQS 7 / Tape B | 224.0.203.38 | 45007 |
| CQS 8 / Tape A | 224.0.203.8 | 45008 | CQS 8 / Tape B | 224.0.203.39 | 45008 |
| CQS 9 / Tape A | 224.0.203.9 | 45009 | CQS 9 / Tape B | 224.0.203.40 | 45009 |
| CQS 10 / Tape A | 224.0.203.10 | 45010 | CQS 10 / Tape B | 224.0.203.41 | 45010 |
| CQS 11 / Tape A | 224.0.203.11 | 45011 | CQS 11 / Tape B | 224.0.203.42 | 45011 |
| CQS 12 / Tape A | 224.0.203.12 | 45012 | CQS 12 / Tape B | 224.0.203.43 | 45012 |

**PRODUCTION "DATA STREAM - B" SUBNETS:**

| 224.0.203.128/28 | | | 224.0.203.160/28 | | |
|---|---|---|---|---|---|
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| CQS 1 / Tape A | 224.0.203.128 | 45001 | CQS 1 / Tape B | 224.0.203.160 | 45001 |
| CQS 2 / Tape A | 224.0.203.129 | 45002 | CQS 2 / Tape B | 224.0.203.161 | 45002 |
| CQS 3 / Tape A | 224.0.203.130 | 45003 | CQS 3 / Tape B | 224.0.203.162 | 45003 |
| CQS 4 / Tape A | 224.0.203.131 | 45004 | CQS 4 / Tape B | 224.0.203.163 | 45004 |
| CQS 5 / Tape A | 224.0.203.132 | 45005 | CQS 5 / Tape B | 224.0.203.164 | 45005 |
| CQS 6 / Tape A | 224.0.203.133 | 45006 | CQS 6 / Tape B | 224.0.203.165 | 45006 |
| CQS 7 / Tape A | 224.0.203.134 | 45007 | CQS 7 / Tape B | 224.0.203.166 | 45007 |
| CQS 8 / Tape A | 224.0.203.135 | 45008 | CQS 8 / Tape B | 224.0.203.167 | 45008 |
| CQS 9 / Tape A | 224.0.203.136 | 45009 | CQS 9 / Tape B | 224.0.203.168 | 45009 |
| CQS 10 / Tape A | 224.0.203.137 | 45010 | CQS 10 / Tape B | 224.0.203.169 | 45010 |
| CQS 11 / Tape A | 224.0.203.138 | 45011 | CQS 11 / Tape B | 224.0.203.170 | 45011 |
| CQS 12 / Tape A | 224.0.203.139 | 45012 | CQS 12 / Tape B | 224.0.203.171 | 45012 |

# Appendix 4 - CQS Retransmission IP Multicast Feeds, Dual Set (Network 'A' / Network 'B')

### MULTICAST DATA: RETRANSMISSION GROUP, DUAL SET

| | |
|---|---|
| PRODUCT NAME: | CQS |
| NETWORK SUBNETS: | TWO (2) NETWORK SUBNETS GROUPS PER DATA STREAM |

**RETRANSMISSION "DATA STREAM - A" SUBNETS:**

| 224.0.203.64/28 | | | 224.0.203.96/28 | | |
|---|---|---|---|---|---|
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| CQS 1 / Tape A | 224.0.203.64 | 46001 | CQS 1 / Tape B | 224.0.203.96 | 46001 |
| CQS 2 / Tape A | 224.0.203.65 | 46002 | CQS 2 / Tape B | 224.0.203.97 | 46002 |
| CQS 3 / Tape A | 224.0.203.66 | 46003 | CQS 3 / Tape B | 224.0.203.98 | 46003 |
| CQS 4 / Tape A | 224.0.203.67 | 46004 | CQS 4 / Tape B | 224.0.203.99 | 46004 |
| CQS 5 / Tape A | 224.0.203.68 | 46005 | CQS 5 / Tape B | 224.0.203.100 | 46005 |
| CQS 6 / Tape A | 224.0.203.69 | 46006 | CQS 6 / Tape B | 224.0.203.101 | 46006 |
| CQS 7 / Tape A | 224.0.203.70 | 46007 | CQS 7 / Tape B | 224.0.203.102 | 46007 |
| CQS 8 / Tape A | 224.0.203.71 | 46008 | CQS 8 / Tape B | 224.0.203.103 | 46008 |
| CQS 9 / Tape A | 224.0.203.72 | 46009 | CQS 9 / Tape B | 224.0.203.104 | 46009 |
| CQS 10 / Tape A | 224.0.203.73 | 46010 | CQS 10 / Tape B | 224.0.203.105 | 46010 |
| CQS 11 / Tape A | 224.0.203.74 | 46011 | CQS 11 / Tape B | 224.0.203.106 | 46011 |
| CQS 12 / Tape A | 224.0.203.75 | 46012 | CQS 12 / Tape B | 224.0.203.107 | 46012 |

**RETRANSMISSION "DATA STREAM - B" SUBNETS:**

| 224.0.203.192/28 | | | 224.0.203.224/28 | | |
|---|---|---|---|---|---|
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| CQS 1 / Tape A | 224.0.203.192 | 46001 | CQS 1 / Tape B | 224.0.203.224 | 46001 |
| CQS 2 / Tape A | 224.0.203.193 | 46002 | CQS 2 / Tape B | 224.0.203.225 | 46002 |
| CQS 3 / Tape A | 224.0.203.194 | 46003 | CQS 3 / Tape B | 224.0.203.226 | 46003 |
| CQS 4 / Tape A | 224.0.203.195 | 46004 | CQS 4 / Tape B | 224.0.203.227 | 46004 |
| CQS 5 / Tape A | 224.0.203.196 | 46005 | CQS 5 / Tape B | 224.0.203.228 | 46005 |
| CQS 6 / Tape A | 224.0.203.197 | 46006 | CQS 6 / Tape B | 224.0.203.229 | 46006 |
| CQS 7 / Tape A | 224.0.203.198 | 46007 | CQS 7 / Tape B | 224.0.203.230 | 46007 |
| CQS 8 / Tape A | 224.0.203.199 | 46008 | CQS 8 / Tape B | 224.0.203.231 | 46008 |
| CQS 9 / Tape A | 224.0.203.200 | 46009 | CQS 9 / Tape B | 224.0.203.232 | 46009 |
| CQS 10 / Tape A | 224.0.203.201 | 46010 | CQS 10 / Tape B | 224.0.203.233 | 46010 |
| CQS 11 / Tape A | 224.0.203.202 | 46011 | CQS 11 / Tape B | 224.0.203.234 | 46011 |
| CQS 12 / Tape A | 224.0.203.203 | 46012 | CQS 12 / Tape B | 224.0.203.235 | 46012 |

# Appendix 4 - CQS Playback Test IP Multicast Feeds, Dual Set (Network 'A' / Network 'B')

**MULTICAST DATA: PLAYBACK TEST GROUP, DUAL SET**

| PRODUCT NAME: | CQS |
|---|---|
| NETWORK SUBNETS: | TWO (2) NETWORK SUBNETS GROUPS PER DATA STREAM |

**PLAYBACK TEST "DATA STREAM - A" SUBNETS:**

| 224.0.200.60/31 | | | 224.0.200.62/31 | | |
|---|---|---|---|---|---|
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| CQS 1 / Tape A | 224.0.200.60 | 47001 | CQS 1 / Tape B | 224.0.200.62 | 47001 |
| CQS 2 / Tape A | 224.0.200.60 | 47002 | CQS 2 / Tape B | 224.0.200.62 | 47002 |
| CQS 3 / Tape A | 224.0.200.60 | 47003 | CQS 3 / Tape B | 224.0.200.62 | 47003 |
| CQS 4 / Tape A | 224.0.200.60 | 47004 | CQS 4 / Tape B | 224.0.200.62 | 47004 |
| CQS 5 / Tape A | 224.0.200.60 | 47005 | CQS 5 / Tape B | 224.0.200.62 | 47005 |
| CQS 6 / Tape A | 224.0.200.60 | 47006 | CQS 6 / Tape B | 224.0.200.62 | 47006 |
| CQS 7 / Tape A | 224.0.200.60 | 47007 | CQS 7 / Tape B | 224.0.200.62 | 47007 |
| CQS 8 / Tape A | 224.0.200.60 | 47008 | CQS 8 / Tape B | 224.0.200.62 | 47008 |
| CQS 9 / Tape A | 224.0.200.60 | 47009 | CQS 9 / Tape B | 224.0.200.62 | 47009 |
| CQS 10 / Tape A | 224.0.200.60 | 47010 | CQS 10 / Tape B | 224.0.200.62 | 47010 |
| CQS 11 / Tape A | 224.0.200.60 | 47011 | CQS 11 / Tape B | 224.0.200.62 | 47011 |
| CQS 12 / Tape A | 224.0.200.60 | 47012 | CQS 12 / Tape B | 224.0.200.62 | 47012 |

**PLAYBACK TEST "DATA STREAM - B" SUBNETS:**

| 224.0.200.188/31 | | | 224.0.200.190/31 | | |
|---|---|---|---|---|---|
| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
| CQS 1 / Tape A | 224.0.200.188 | 47001 | CQS 1 / Tape B | 224.0.200.190 | 47001 |
| CQS 2 / Tape A | 224.0.200.188 | 47002 | CQS 2 / Tape B | 224.0.200.190 | 47002 |
| CQS 3 / Tape A | 224.0.200.188 | 47003 | CQS 3 / Tape B | 224.0.200.190 | 47003 |
| CQS 4 / Tape A | 224.0.200.188 | 47004 | CQS 4 / Tape B | 224.0.200.190 | 47004 |
| CQS 5 / Tape A | 224.0.200.188 | 47005 | CQS 5 / Tape B | 224.0.200.190 | 47005 |
| CQS 6 / Tape A | 224.0.200.188 | 47006 | CQS 6 / Tape B | 224.0.200.190 | 47006 |
| CQS 7 / Tape A | 224.0.200.188 | 47007 | CQS 7 / Tape B | 224.0.200.190 | 47007 |
| CQS 8 / Tape A | 224.0.200.188 | 47008 | CQS 8 / Tape B | 224.0.200.190 | 47008 |
| CQS 9 / Tape A | 224.0.200.188 | 47009 | CQS 9 / Tape B | 224.0.200.190 | 47009 |
| CQS 10 / Tape A | 224.0.200.188 | 47010 | CQS 10 / Tape B | 224.0.200.190 | 47010 |
| CQS 11 / Tape A | 224.0.200.188 | 47011 | CQS 11 / Tape B | 224.0.200.190 | 47011 |
| CQS 12 / Tape A | 224.0.200.188 | 47012 | CQS 12 / Tape B | 224.0.200.190 | 47012 |

# Appendix 4 - CQS Snapshot Retransmission IP Multicast Feeds, Dual Set (Network 'A' / Network 'B')

**MULTICAST DATA: SNAPSHOT RETRANSMISSION GROUP, DUAL SET**

| PRODUCT NAME: | CQS |
|---|---|
| NETWORK SUBNETS: | TWO (2) NETWORK SUBNETS GROUPS PER DATA STREAM |

**SNAPSHOT RETRANSMISSION "DATA STREAM - A" SUBNETS:**

| 224.0.201.0/27 | 224.0.201.32/27 |
|---|---|

| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|---|---|---|
| CQS 1 / Tape A | 224.0.201.1 | 48001 | CQS 1 / Tape B | 224.0.201.32 | 48001 |
| CQS 2 / Tape A | 224.0.201.2 | 48002 | CQS 2 / Tape B | 224.0.201.33 | 48002 |
| CQS 3 / Tape A | 224.0.201.3 | 48003 | CQS 3 / Tape B | 224.0.201.34 | 48003 |
| CQS 4 / Tape A | 224.0.201.4 | 48004 | CQS 4 / Tape B | 224.0.201.35 | 48004 |
| CQS 5 / Tape A | 224.0.201.5 | 48005 | CQS 5 / Tape B | 224.0.201.36 | 48005 |
| CQS 6 / Tape A | 224.0.201.6 | 48006 | CQS 6 / Tape B | 224.0.201.37 | 48006 |
| CQS 7 / Tape A | 224.0.201.7 | 48007 | CQS 7 / Tape B | 224.0.201.38 | 48007 |
| CQS 8 / Tape A | 224.0.201.8 | 48008 | CQS 8 / Tape B | 224.0.201.39 | 48008 |
| CQS 9 / Tape A | 224.0.201.9 | 48009 | CQS 9 / Tape B | 224.0.201.40 | 48009 |
| CQS 10 / Tape A | 224.0.201.10 | 48010 | CQS 10 / Tape B | 224.0.201.41 | 48010 |
| CQS 11 / Tape A | 224.0.201.11 | 48011 | CQS 11 / Tape B | 224.0.201.42 | 48011 |
| CQS 12 / Tape A | 224.0.201.12 | 48012 | CQS 12 / Tape B | 224.0.201.43 | 48012 |

**SNAPSHOT RETRANSMISSION "DATA STREAM - B" SUBNETS:**

| 224.0.201.128/27 | 224.0.201.160/27 |
|---|---|

| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|---|---|---|
| CQS 1 / Tape A | 224.0.201.128 | 48001 | CQS 1 / Tape B | 224.0.201.160 | 48001 |
| CQS 2 / Tape A | 224.0.201.129 | 48002 | CQS 2 / Tape B | 224.0.201.161 | 48002 |
| CQS 3 / Tape A | 224.0.201.130 | 48003 | CQS 3 / Tape B | 224.0.201.162 | 48003 |
| CQS 4 / Tape A | 224.0.201.131 | 48004 | CQS 4 / Tape B | 224.0.201.163 | 48004 |
| CQS 5 / Tape A | 224.0.201.132 | 48005 | CQS 5 / Tape B | 224.0.201.164 | 48005 |
| CQS 6 / Tape A | 224.0.201.133 | 48006 | CQS 6 / Tape B | 224.0.201.165 | 48006 |
| CQS 7 / Tape A | 224.0.201.134 | 48007 | CQS 7 / Tape B | 224.0.201.166 | 48007 |
| CQS 8 / Tape A | 224.0.201.135 | 48008 | CQS 8 / Tape B | 224.0.201.167 | 48008 |
| CQS 9 / Tape A | 224.0.201.136 | 48009 | CQS 9 / Tape B | 224.0.201.168 | 48009 |
| CQS 10 / Tape A | 224.0.201.137 | 48010 | CQS 10 / Tape B | 224.0.201.169 | 48010 |
| CQS 11 / Tape A | 224.0.201.138 | 48011 | CQS 11 / Tape B | 224.0.201.170 | 48011 |
| CQS 12 / Tape A | 224.0.201.139 | 48012 | CQS 12 / Tape B | 224.0.201.171 | 48012 |

# 10 Appendix 5 - CTS Production, Real-Time IP Multicast Feeds, Dual Sets (Network 'A' / Network 'B')

**MULTICAST DATA: PRODUCTION "A-STREAM" & PRODUCTION "B-STREAM"**

| PRODUCT NAME: | CTS |
|---|---|
| NETWORK SUBNETS: | THREE (3) NETWORK SUBNETS GROUPS PER DATA STREAM |

**PRODUCTION "DATA STREAM - A" SUBNETS:**

| 224.0.202.0/28 | 224.0.202.32/28 | 224.0.200.96/30 |
|---|---|---|

| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|---|---|---|
| CTS 1 / Tape A | 224.0.202.1 | 45001 | CTS 1 / Tape B | 224.0.202.32 | 45001 |
| CTS 2 / Tape A | 224.0.202.2 | 45002 | CTS 2 / Tape B | 224.0.202.33 | 45002 |
| CTS 3 / Tape A | 224.0.202.3 | 45003 | CTS 3 / Tape B | 224.0.202.34 | 45003 |
| CTS 4 / Tape A | 224.0.202.4 | 45004 | CTS 4 / Tape B | 224.0.202.35 | 45004 |
| CTS 5 / Tape A | 224.0.202.5 | 45005 | CTS 5 / Tape B | 224.0.202.36 | 45005 |
| CTS 6 / Tape A | 224.0.202.6 | 45006 | CTS 6 / Tape B | 224.0.202.37 | 45006 |
| CTS 7 / Tape A | 224.0.202.7 | 45007 | CTS 7 / Tape B | 224.0.202.38 | 45007 |
| CTS 8 / Tape A | 224.0.202.8 | 45008 | CTS 8 / Tape B | 224.0.202.39 | 45008 |
| CTS 9 / Tape A | 224.0.202.9 | 45009 | CTS 9 / Tape B | 224.0.202.40 | 45009 |
| CTS 10 / Tape A | 224.0.202.10 | 45010 | CTS 10 / Tape B | 224.0.202.41 | 45010 |
| CTS 11 / Tape A | 224.0.202.11 | 45011 | CTS 11 / Tape B | 224.0.202.42 | 45011 |
| CTS 12 / Tape A | 224.0.202.12 | 45012 | CTS 12 / Tape B | 224.0.202.43 | 45012 |
| | | | | | |
| | | | CTS INDEX 1 / Tape B | 224.0.200.96 | 45001 |
| | | | CTS INDEX 2 / Tape B | 224.0.200.97 | 45002 |

**PRODUCTION "DATA STREAM - B" SUBNETS:**

| 224.0.202.128/28 | 224.0.202.160/28 | 224.0.200.224/30 |
|---|---|---|

| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|---|---|---|
| CTS 1 / Tape A | 224.0.202.128 | 45001 | CTS 1 / Tape B | 224.0.202.160 | 45001 |
| CTS 2 / Tape A | 224.0.202.129 | 45002 | CTS 2 / Tape B | 224.0.202.161 | 45002 |
| CTS 3 / Tape A | 224.0.202.130 | 45003 | CTS 3 / Tape B | 224.0.202.162 | 45003 |
| CTS 4 / Tape A | 224.0.202.131 | 45004 | CTS 4 / Tape B | 224.0.202.163 | 45004 |
| CTS 5 / Tape A | 224.0.202.132 | 45005 | CTS 5 / Tape B | 224.0.202.164 | 45005 |
| CTS 6 / Tape A | 224.0.202.133 | 45006 | CTS 6 / Tape B | 224.0.202.165 | 45006 |
| CTS 7 / Tape A | 224.0.202.134 | 45007 | CTS 7 / Tape B | 224.0.202.166 | 45007 |
| CTS 8 / Tape A | 224.0.202.135 | 45008 | CTS 8 / Tape B | 224.0.202.167 | 45008 |
| CTS 9 / Tape A | 224.0.202.136 | 45009 | CTS 9 / Tape B | 224.0.202.168 | 45009 |
| CTS 10 / Tape A | 224.0.202.137 | 45010 | CTS 10 / Tape B | 224.0.202.169 | 45010 |
| CTS 11 / Tape A | 224.0.202.138 | 45011 | CTS 11 / Tape B | 224.0.202.170 | 45011 |
| CTS 12 / Tape A | 224.0.202.139 | 45012 | CTS 12 / Tape B | 224.0.202.171 | 45012 |
| | | | | | |
| | | | CTS INDEX 1 / Tape B | 224.0.200.224 | 45001 |
| | | | CTS INDEX 2 / Tape B | 224.0.200.225 | 45002 |

# Appendix 5 - CTS Retransmission IP Multicast Feeds, Dual Set (Network 'A' / Network 'B')

**MULTICAST DATA: RETRANSMISSION GROUP, DUAL SET**

| PRODUCT NAME: | CTS |
|---|---|
| NETWORK SUBNETS: | THREE (3) NETWORK SUBNETS GROUPS PER DATA STREAM |

**RETRANSMISSION "DATA STREAM - A" SUBNETS:**

| 224.0.202.64/28 | 224.0.202.96/28 | 224.0.200.100/30 |
|---|---|---|

| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|---|---|---|
| CTS 1 / Tape A | 224.0.202.64 | 46001 | CTS 1 / Tape B | 224.0.202.96 | 46001 |
| CTS 2 / Tape A | 224.0.202.65 | 46002 | CTS 2 / Tape B | 224.0.202.97 | 46002 |
| CTS 3 / Tape A | 224.0.202.66 | 46003 | CTS 3 / Tape B | 224.0.202.98 | 46003 |
| CTS 4 / Tape A | 224.0.202.67 | 46004 | CTS 4 / Tape B | 224.0.202.99 | 46004 |
| CTS 5 / Tape A | 224.0.202.68 | 46005 | CTS 5 / Tape B | 224.0.202.100 | 46005 |
| CTS 6 / Tape A | 224.0.202.69 | 46006 | CTS 6 / Tape B | 224.0.202.101 | 46006 |
| CTS 7 / Tape A | 224.0.202.70 | 46007 | CTS 7 / Tape B | 224.0.202.102 | 46007 |
| CTS 8 / Tape A | 224.0.202.71 | 46008 | CTS 8 / Tape B | 224.0.202.103 | 46008 |
| CTS 9 / Tape A | 224.0.202.72 | 46009 | CTS 9 / Tape B | 224.0.202.104 | 46009 |
| CTS 10 / Tape A | 224.0.202.73 | 46010 | CTS 10 / Tape B | 224.0.202.105 | 46010 |
| CTS 11 / Tape A | 224.0.202.74 | 46011 | CTS 11 / Tape B | 224.0.202.106 | 46011 |
| CTS 12 / Tape A | 224.0.202.75 | 46012 | CTS 12 / Tape B | 224.0.202.107 | 46012 |
| | | | | | |
| | | | CTS INDEX 1 / Tape B | 224.0.200.100 | 46001 |
| | | | CTS INDEX 2 / Tape B | 224.0.200.101 | 46002 |

**RETRANSMISSION "DATA STREAM - B" SUBNETS:**

| 224.0.202.192/28 | 224.0.202.224/28 | 224.0.200.228/30 |
|---|---|---|

| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|---|---|---|
| CTS 1 / Tape A | 224.0.202.192 | 46001 | CTS 1 / Tape B | 224.0.202.224 | 46001 |
| CTS 2 / Tape A | 224.0.202.193 | 46002 | CTS 2 / Tape B | 224.0.202.225 | 46002 |
| CTS 3 / Tape A | 224.0.202.194 | 46003 | CTS 3 / Tape B | 224.0.202.226 | 46003 |
| CTS 4 / Tape A | 224.0.202.195 | 46004 | CTS 4 / Tape B | 224.0.202.227 | 46004 |
| CTS 5 / Tape A | 224.0.202.196 | 46005 | CTS 5 / Tape B | 224.0.202.228 | 46005 |
| CTS 6 / Tape A | 224.0.202.197 | 46006 | CTS 6 / Tape B | 224.0.202.229 | 46006 |
| CTS 7 / Tape A | 224.0.202.198 | 46007 | CTS 7 / Tape B | 224.0.202.230 | 46007 |
| CTS 8 / Tape A | 224.0.202.199 | 46008 | CTS 8 / Tape B | 224.0.202.231 | 46008 |
| CTS 9 / Tape A | 224.0.202.200 | 46009 | CTS 9 / Tape B | 224.0.202.232 | 46009 |
| CTS 10 / Tape A | 224.0.202.201 | 46010 | CTS 10 / Tape B | 224.0.202.233 | 46010 |
| CTS 11 / Tape A | 224.0.202.202 | 46011 | CTS 11 / Tape B | 224.0.202.234 | 46011 |
| CTS 12 / Tape A | 224.0.202.203 | 46012 | CTS 12 / Tape B | 224.0.202.235 | 46012 |
| | | | | | |
| | | | CTS INDEX 1 / Tape B | 224.0.200.228 | 46001 |
| | | | CTS INDEX 2 / Tape B | 224.0.200.229 | 46002 |

# Appendix 5 - CTS Playback Test IP Multicast Feeds, Dual Sets (Network 'A' / Network 'B')

**MULTICAST DATA: PLAYBACK TEST GROUP, DUAL SET**

| PRODUCT NAME: | CTS |
|---|---|
| NETWORK SUBNETS: | THREE (3) NETWORK SUBNETS GROUPS PER DATA STREAM |

**PLAYBACK TEST "DATA STREAM - A" SUBNETS:**

| 224.0.200.56/31 | 224.0.200.58/31 | 224.0.200.104/30 |
|---|---|---|

| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|---|---|---|
| CTS 1 / Tape A | 224.0.200.56 | 47001 | CTS 1 / Tape B | 224.0.200.58 | 47001 |
| CTS 2 / Tape A | 224.0.200.56 | 47002 | CTS 2 / Tape B | 224.0.200.58 | 47002 |
| CTS 3 / Tape A | 224.0.200.56 | 47003 | CTS 3 / Tape B | 224.0.200.58 | 47003 |
| CTS 4 / Tape A | 224.0.200.56 | 47004 | CTS 4 / Tape B | 224.0.200.58 | 47004 |
| CTS 5 / Tape A | 224.0.200.56 | 47005 | CTS 5 / Tape B | 224.0.200.58 | 47005 |
| CTS 6 / Tape A | 224.0.200.56 | 47006 | CTS 6 / Tape B | 224.0.200.58 | 47006 |
| CTS 7 / Tape A | 224.0.200.56 | 47007 | CTS 7 / Tape B | 224.0.200.58 | 47007 |
| CTS 8 / Tape A | 224.0.200.56 | 47008 | CTS 8 / Tape B | 224.0.200.58 | 47008 |
| CTS 9 / Tape A | 224.0.200.56 | 47009 | CTS 9 / Tape B | 224.0.200.58 | 47009 |
| CTS 10 / Tape A | 224.0.200.56 | 47010 | CTS 10 / Tape B | 224.0.200.58 | 47010 |
| CTS 11 / Tape A | 224.0.200.56 | 47011 | CTS 11 / Tape B | 224.0.200.58 | 47011 |
| CTS 12 / Tape A | 224.0.200.56 | 47012 | CTS 12 / Tape B | 224.0.200.58 | 47012 |
| | | | | | |
| | | | CTS INDEX 1 / Tape B | 224.0.200.104 | 47001 |
| | | | CTS INDEX 2 / Tape B | 224.0.200.105 | 47002 |

**PLAYBACK TEST "DATA STREAM - B" SUBNETS:**

| 224.0.200.184/31 | 224.0.200.186/31 | 224.0.200.232/30 |
|---|---|---|

| Originated Data Line | Multicast Group Address | Multicast Group Port | Originated Data Line | Multicast Group Address | Multicast Group Port |
|---|---|---|---|---|---|
| CTS 1 / Tape A | 224.0.200.184 | 47001 | CTS 1 / Tape B | 224.0.200.186 | 47001 |
| CTS 2 / Tape A | 224.0.200.184 | 47002 | CTS 2 / Tape B | 224.0.200.186 | 47002 |
| CTS 3 / Tape A | 224.0.200.184 | 47003 | CTS 3 / Tape B | 224.0.200.186 | 47003 |
| CTS 4 / Tape A | 224.0.200.184 | 47004 | CTS 4 / Tape B | 224.0.200.186 | 47004 |
| CTS 5 / Tape A | 224.0.200.184 | 47005 | CTS 5 / Tape B | 224.0.200.186 | 47005 |
| CTS 6 / Tape A | 224.0.200.184 | 47006 | CTS 6 / Tape B | 224.0.200.186 | 47006 |
| CTS 7 / Tape A | 224.0.200.184 | 47007 | CTS 7 / Tape B | 224.0.200.186 | 47007 |
| CTS 8 / Tape A | 224.0.200.184 | 47008 | CTS 8 / Tape B | 224.0.200.186 | 47008 |
| CTS 9 / Tape A | 224.0.200.184 | 47009 | CTS 9 / Tape B | 224.0.200.186 | 47009 |
| CTS 10 / Tape A | 224.0.200.184 | 47010 | CTS 10 / Tape B | 224.0.200.186 | 47010 |
| CTS 11 / Tape A | 224.0.200.184 | 47011 | CTS 11 / Tape B | 224.0.200.186 | 47011 |
| CTS 12 / Tape A | 224.0.200.184 | 47012 | CTS 12 / Tape B | 224.0.200.186 | 47012 |
| | | | | | |
| | | | CTS INDEX 1 / Tape B | 224.0.200.232 | 47001 |
| | | | CTS INDEX 2 / Tape B | 224.0.200.233 | 47002 |